

SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone Presentation

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Project Context & Scope

- **Core Goal:** Predict whether the Falcon 9 first stage booster will land successfully.
- **Reusable Systems:** Recovering first stages saves SpaceX millions per launch (~\$62M vs competitors' \$165M+).
- **Strategic Value:** Allows competitors to estimate launch bidding costs and insurances.

Executive Summary: Objective & Method

- **Ingestion Pipeline:** Cleaned API response data and scraped flight tables into 90 records.
- **GIS Proximity:** Analyzed launch site coordinates relative to coastlines and transportation facilities.
- **Machine Learning:** Tuned 4 classifiers using GridSearchCV to optimize landing predictions.

The Business Problem: Reusable Launch Systems

- **Capital Conservation:** Rocket engine recovery preserves ~70% of production value.
- **Pricing Disruption:** SpaceX offers launch pricing 60%+ lower than conventional expendable providers.
- **Prediction Utility:** Quantifying landing risk improves business intelligence.

Data Ingestion Pipeline

- **REST API Retrieval:** Extracted flight records and mapped nested resource details.
- **Wikipedia Web Scraping:** Parsed launch tables to cross-reference data.
- **Data Consolidation:** Normalization in Pandas and export to SQLite.

Data Wrangling & Imputation

- **Filtering:** Excluded early Falcon 1 flights to establish a uniform 90-flight Falcon 9 dataset.
- **Imputation:** Calculated mean payload mass (13,104.96 kg) to impute missing rows.
- **Target Labeling:** Engineered binary class variables based on descriptive landing text outcomes.

Chart 1: Flight Number vs Launch Site

- **Observation:** Early launches are centered at CCAFS LC-40.
- **Technology Maturity:** Launch successes (blue dots) dominate late flights across all locations.
- **Strategic Insight:** Confirms improvement in recovery procedures over time.

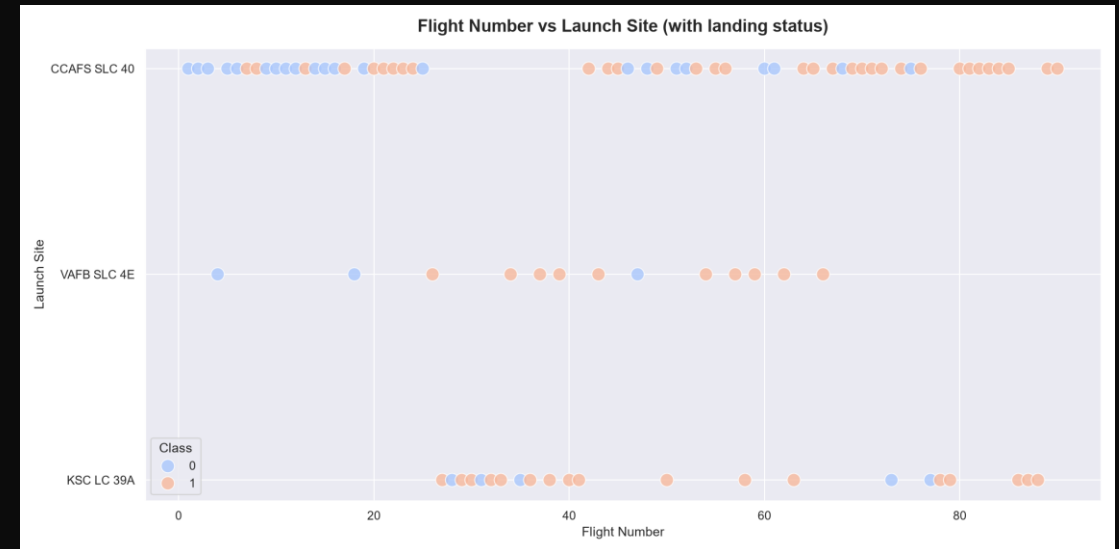


Chart 2: Payload Mass vs Launch Site

- **Observation:** Heavy payloads (above 10,000 kg) are launched from Florida sites.
- **Site Restrictions:** VAFB SLC-4E payloads are consistently capped below 10,000 kg.
- **Recovery Rationale:** Heavy launches show stable landing success records on later flights.

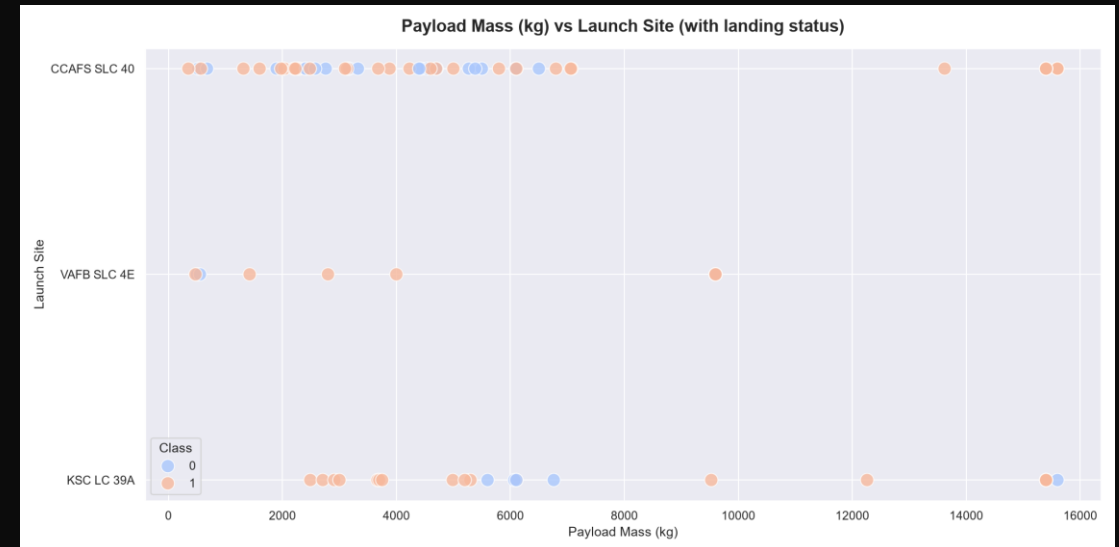
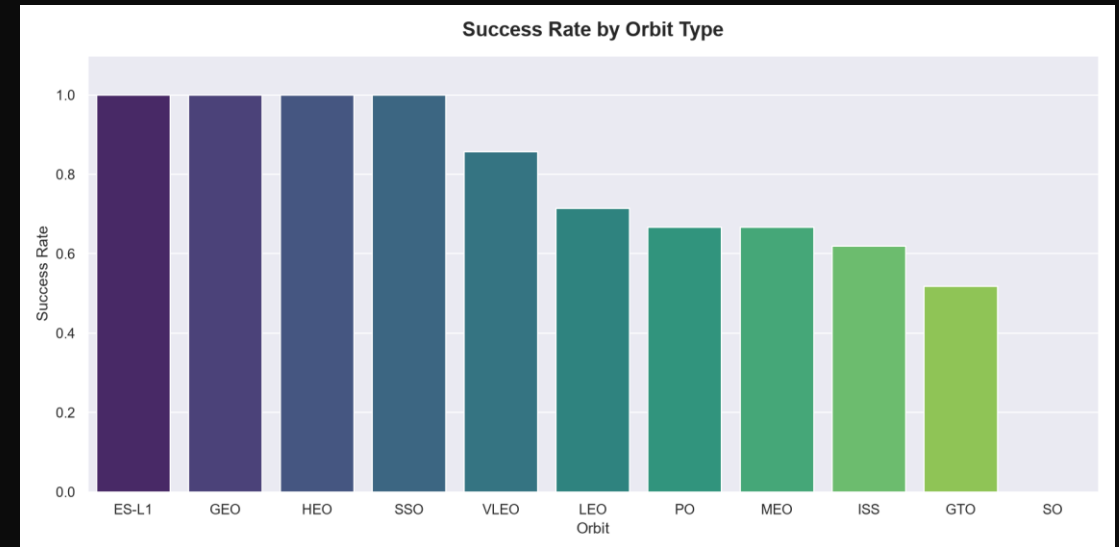


Chart 3: Success Rate by Orbit Type

- **High Success Orbits:** ES-L1, GEO, HEO, and SSO maintain a 100% success rate.
- **High Energy Orbits:** GTO has lower landing success (~50%).
- **Delta-v limitations:** Higher entry speeds leave less fuel for recovery burns.



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- The scatter plot displays the distribution of flight numbers across various orbit types, categorized by two classes. The Y-axis lists orbit types from top to bottom: LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, and GEO. The X-axis represents the Flight Number, ranging from 0 to 90. Class 0 is represented by blue dots, and Class 1 is represented by orange dots.
- Key observations from the plot:
- LEO:** Class 0 flights are concentrated at very low numbers (1-5). Class 1 flights are scattered across the range (10, 18, 30, 38, 45).
 - ISS:** Both classes have flights, with Class 0 appearing at lower numbers (5, 12, 14, 16, 18, 48, 52) and Class 1 at higher numbers (10, 20, 25, 32, 38, 42, 62, 64, 68, 70, 78, 80).
 - PO:** Class 0 flights are at 5 and 48. Class 1 flights are at 10, 28, 35, 38, 42, 50, and 62.
 - GTO:** Both classes have a high density of flights between 5 and 60. Class 0 is more frequent at lower numbers, while Class 1 is more frequent at higher numbers in this range.
 - ES-L1:** Only one Class 1 flight is present at flight number 14.
 - SSO:** Class 1 flights are at 38, 58, 60, 68, and 85.
 - HEO:** One Class 1 flight is at flight number 48.
 - MEO:** Class 0 has one flight at 62. Class 1 has flights at 82 and 90.
 - VLEO:** Both classes have flights, with Class 0 at 75, 76, 77, 78, and 79, and Class 1 at 65, 68, 70, 72, 74, 80, 81, 82, 83, 84, 86, 87, 88, and 89.
 - SO:** One Class 0 flight is at flight number 72.
 - GEO:** One Class 1 flight is at flight number 83.

Chart 5: Payload Mass vs Orbit Type

- **Mass clusters:** Polar orbits have payload weights clustered near 9,600 kg.
- **GTO distribution:** Heavy commercial telecom payloads range from 3,000 to 6,000 kg.
- **LEO variation:** Low-Earth orbit launches carry payload masses up to 15,600 kg.

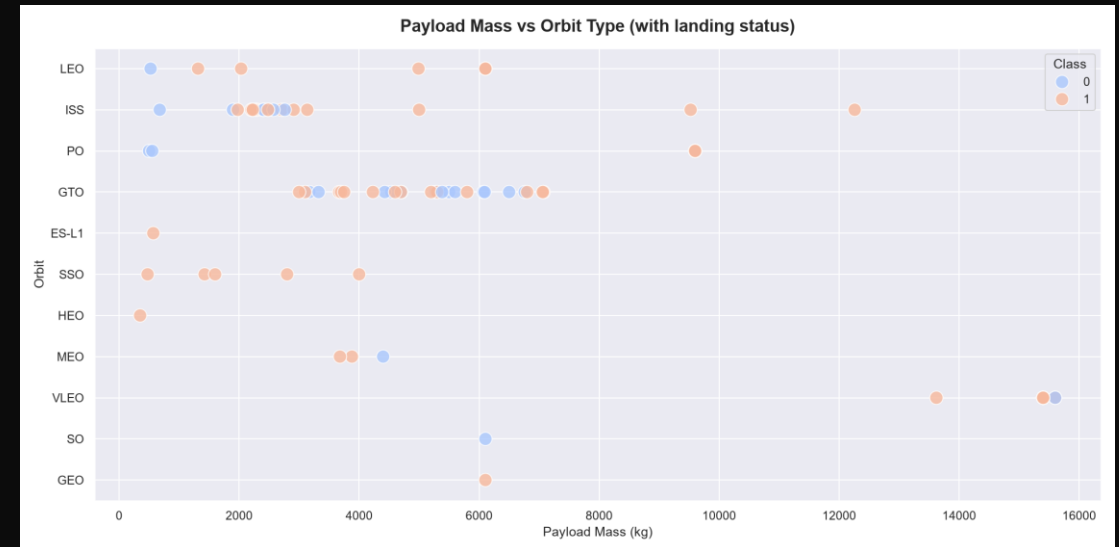


Chart 6: Launch Success Yearly Trend

- **Trend Line:** Rapid growth in success rates after 2015.
- **Stabilization:** Exceeds 80% success rate by 2020.
- **Verification:** Confirms booster reuse is economically and technically viable.

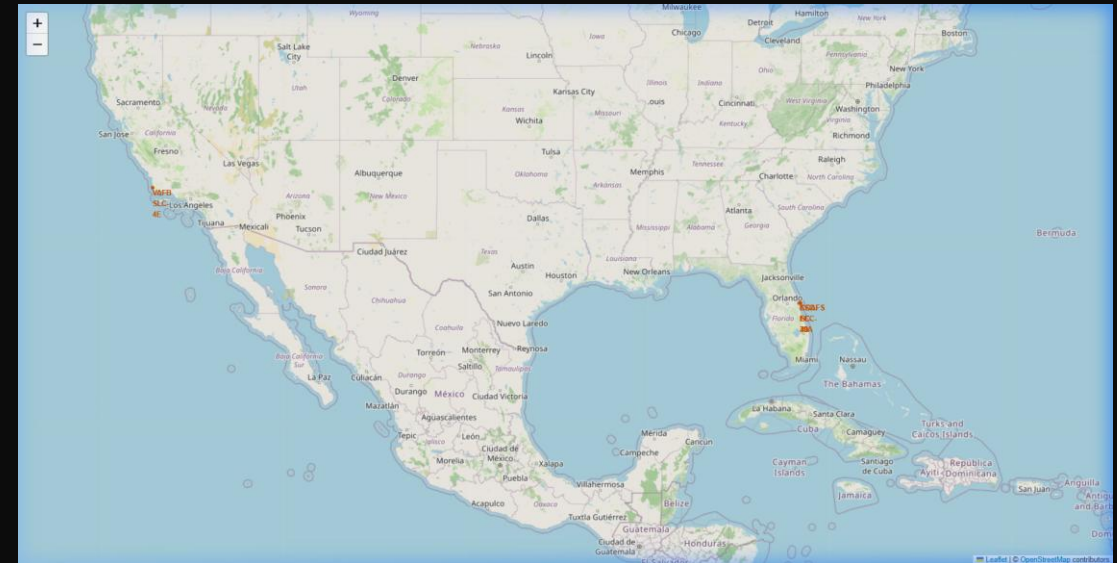


SQL Queries: Auditing the Launch Database

- **Query 1:** Verified CCAFS, KSC, and VAFB launch pad configurations.
- **NASA CRS contracts:** Delivered a total of 45,596 kg to the ISS.
- **Booster versions:** Average payload for F9 v1.1 is 2,928.4 kg.
- **Max Payload:** Starlink flights carry maximum payloads of 15,600 kg.

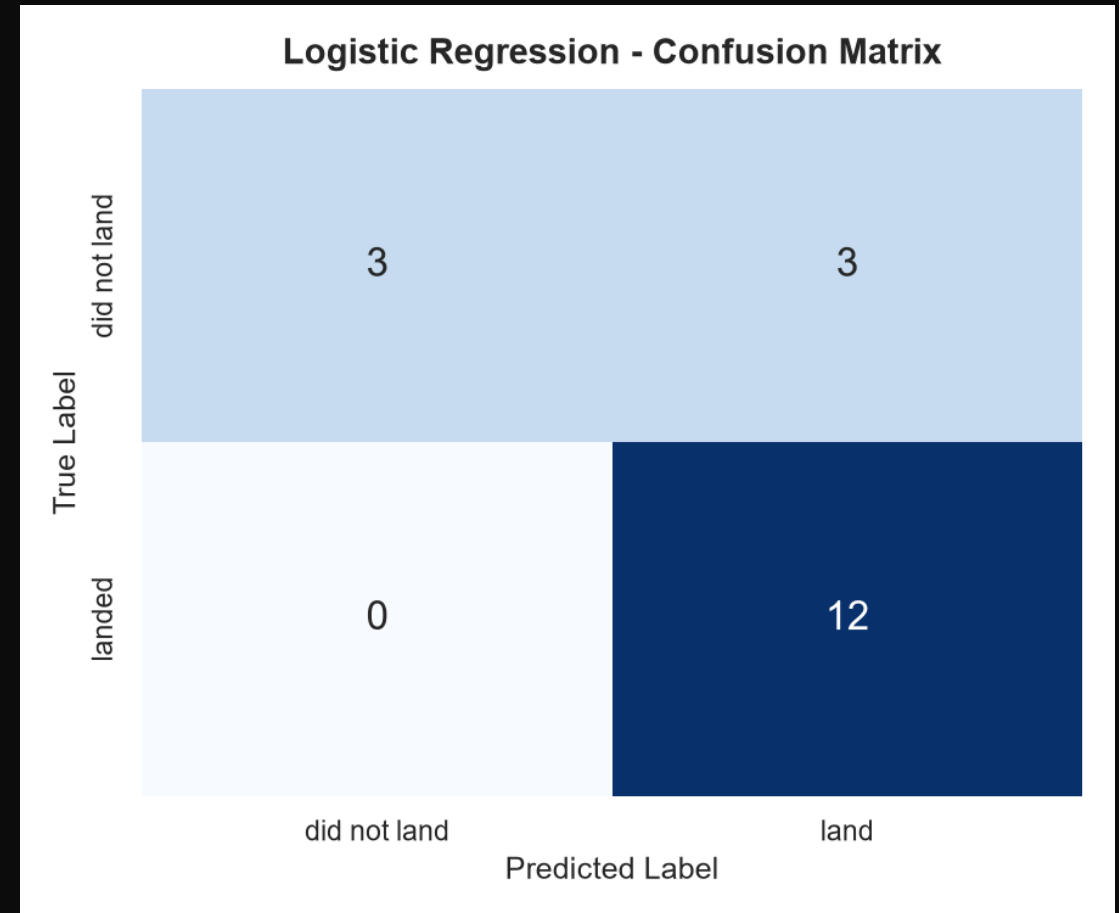
Geographical Mapping & Proximity Heuristics

- **Coastline distance:** 0.58 KM. Placed close to oceans for overwater trajectories.
- **Highway/Railway proximity:** Located within 1.3 KM to facilitate heavy logistics transport.
- **City distance:** 18.0 KM. Isolated from residential populations for safety.



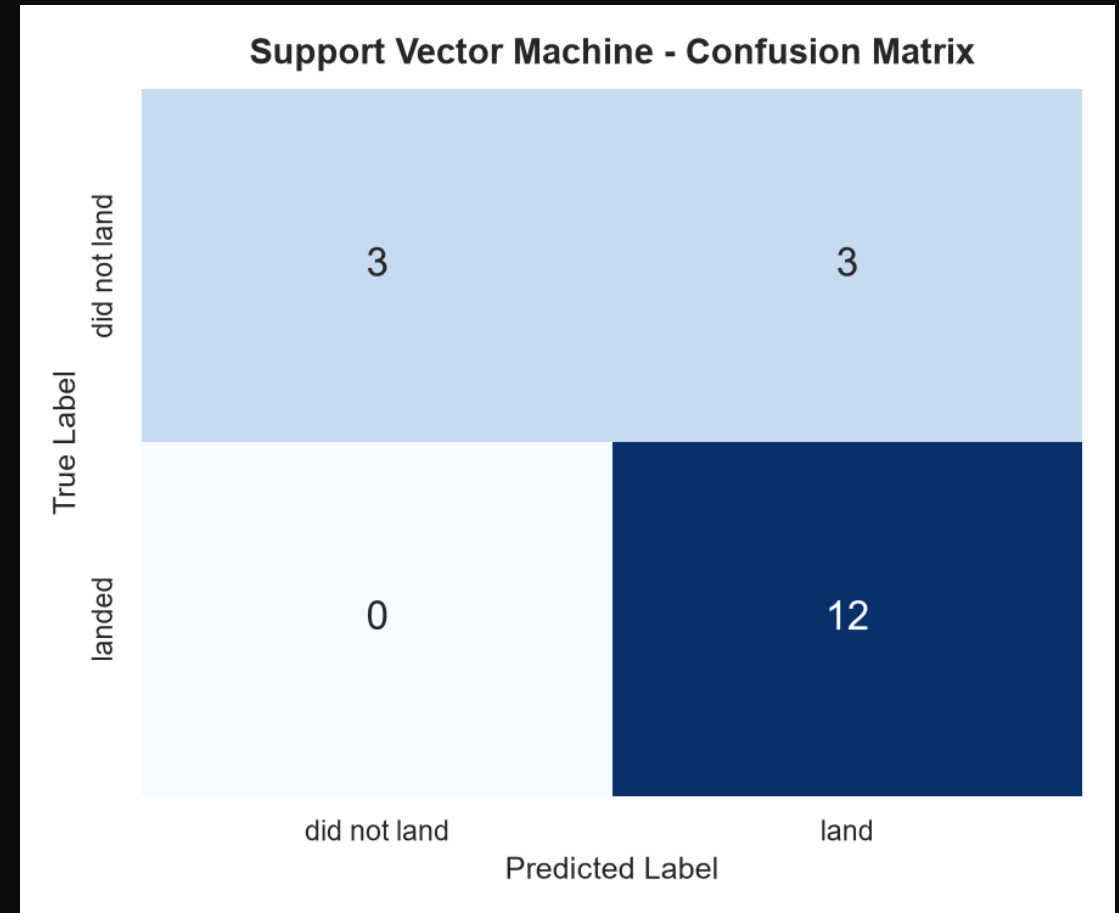
Classifier 1: Logistic Regression

- **Tuning Grid:** Regularization C and L2 solver optimization.
- **Best Parameters:** {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}.
- **Performance:** Test Accuracy of 83.33%.



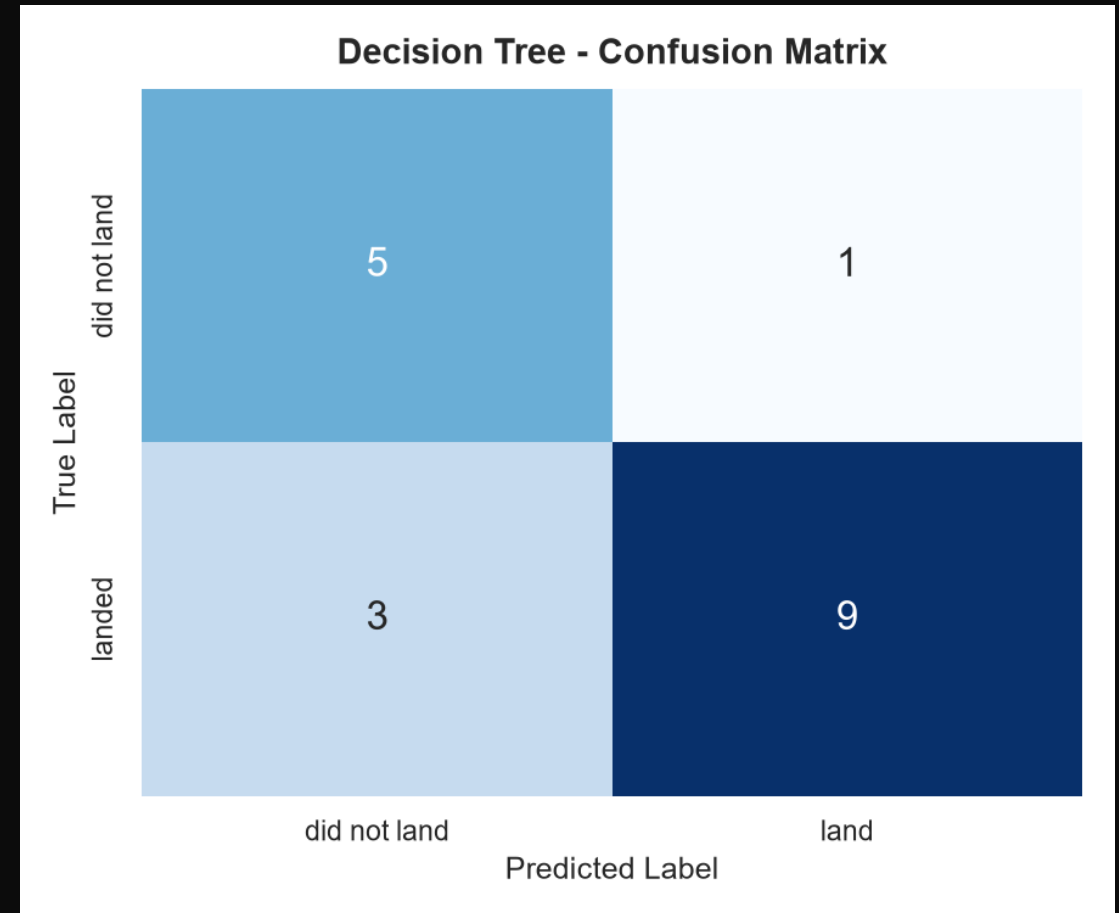
Classifier 2: Support Vector Machine (SVM)

- **Tuning Grid:** Regularization, kernel (linear, rbf, sigmoid) and gamma scaling.
- **Best Parameters:** {'C': 1.0, 'gamma': 0.0316, 'kernel': 'sigmoid'}.
- **Performance:** Test Accuracy of 83.33%.



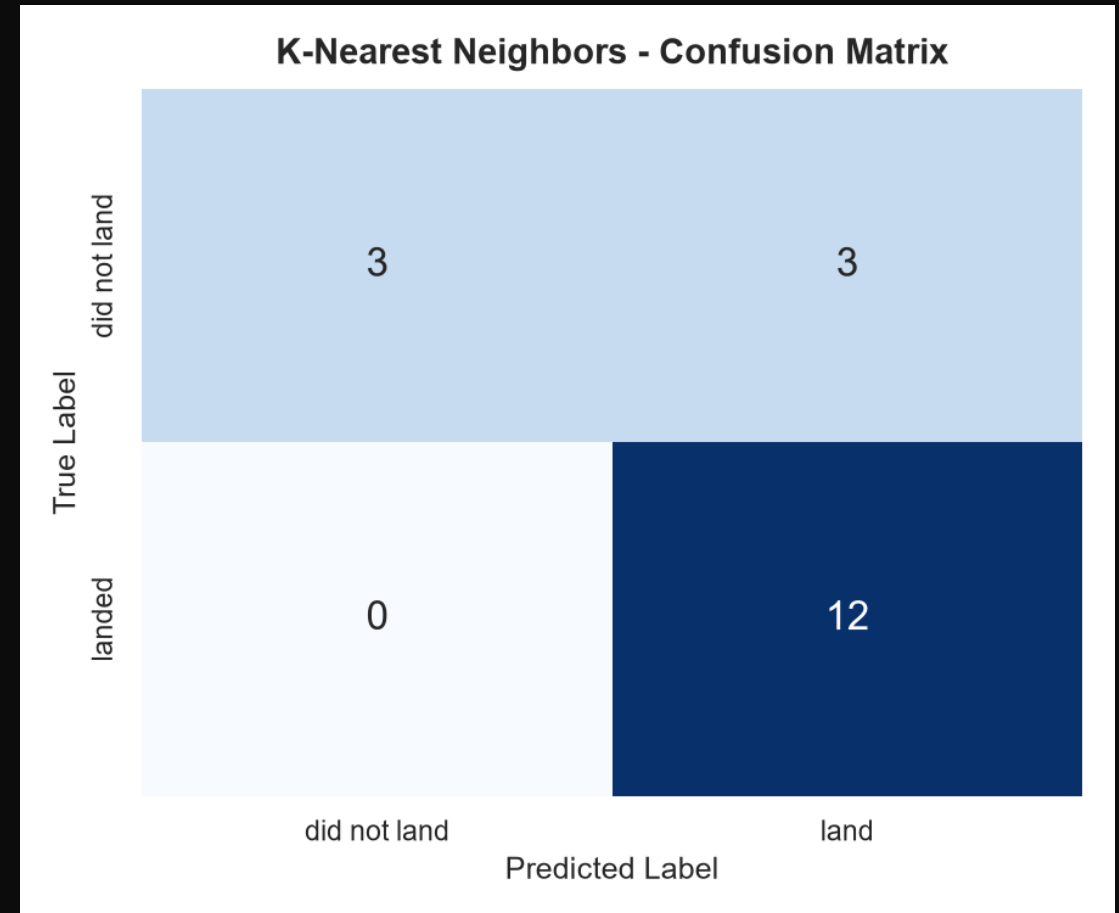
Classifier 3: Decision Tree Classifier

- **Tuning Grid:** Criterion, max depth, splitter, and sample leaf limits.
- **Best Parameters:** {'criterion': 'gini', 'max_depth': 6, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'random'}.
- **Performance:** Test Accuracy of 77.78% (indicates overfitting).



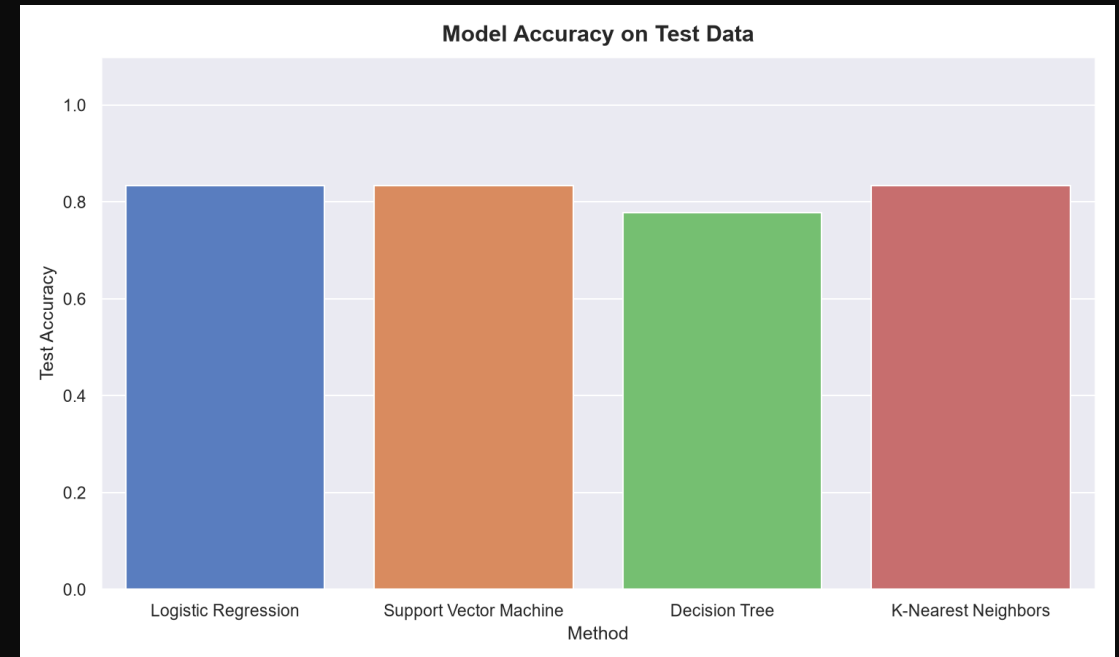
Classifier 4: K-Nearest Neighbors (KNN)

- **Tuning Grid:** Neighbor size, search algorithm, and distance metric.
- **Best Parameters:** {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}.
- **Performance:** Test Accuracy of 83.33%.



Model Accuracy Performance Comparison

- **Validation Accuracy:** Decision Tree scored highest on training folds (87.50%).
- **Test Performance:** LR, SVM, and KNN tied at 83.33% accuracy.
- **Selection:** SVM is selected due to its optimal margin boundaries.



CONCLUSION

Strategic Recommendations

- **Bidding Strategy:** Leverage recovery probabilities to adjust launch bids.
- **Insurance Risk:** Adjust GTO premiums due to lower landing success margins (~50%).
- **Future Enhancements:** Incorporate atmospheric wind telemetry to refine models.